

Who to Call, When to Call: A Data-Driven Framework for Term Deposit Marketing Success

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Note: This document contains a selected excerpt of the full report. If you would like to read the complete analysis, findings, and supporting visualisations, feel free to get in touch; I'd be happy to share the complete version upon request.

1. Executive Summary

This report examines a Portuguese bank's term-deposit telemarketing campaign to identify which customer groups, call strategies, and timing decisions create profitable growth. The analysis uses 45,211 customer contacts and combines bivariate profiling, logistic regression, interaction effects, and an EMV-based decision framework. The main result is clear: campaign execution variables explain subscription behavior far better than demographics alone, but profitability ultimately depends on both conversion and customer value. Only the Not-Labor-Force 60+ segment remains consistently profitable under both standard and balance-adjusted EMV, making it the strongest candidate for expensive outbound calling. The report recommends reallocating calls and budgets towards profitable segments, redesigning campaigns for unprofitable groups, and tightening contact rules around the most responsive periods.

2. Project Goals

The central question is how campaign performance differs across sociodemographic groups in term-deposit marketing. Four research questions guide the study: how demographic factors relate to subscription propensity, whether campaign variables outperform demographics in prediction, whether campaign tactics work differently across customer segments, and which segments remain worth targeting once acquisition cost and customer value are considered.

3. Brief Literature Context

Prior research on the Portuguese bank-marketing dataset treats term-deposit response as a binary classification problem and shows that campaign variables often dominate demographic ones in predictive power. At the same time, age and life-stage remain important because financial needs and savings preferences change over the life cycle. This study extends that work by linking segmentation, prediction, and profitability through Expected Monetary Value (EMV), rather than relying on response rate alone.

4. Methodology

The analysis uses 45,211 customer contacts and 17 variables covering demographics, financial position, and campaign history. Two derived variables were created, specified in Table 1: Age Band and Job Group. The workflow included chi-square tests for bivariate associations, three logistic regression models for predictive comparison, two interaction models to test segment-specific campaign effects, and an EMV framework to assess financial viability.

Table 1: Key Model Inputs

Component	Categories / Variables
Age Band	18–29, 30–39, 40–49, 50–59, 60+
Job Group	White-Collar, Blue-Collar, Self-Employed, Not-Labor-Force
Predictive Models	Demographics only, Campaign only, Combined
Decision Metric	Expected Monetary Value (standard and balance-adjusted)

5. Findings & Discussions

The Demographic Profile: Bivariate Analysis (RQ1)

Age and job group both show statistically significant differences in subscription rates. The 60+ group has the highest conversion rate at 33.6%, while the Not-Labor-Force segment leads all job groups at 21.9%. However, the largest absolute number of subscribers comes from the 30–39 and White-Collar groups because those segments are much larger, not because they are the most responsive.

Table 2: Subscription Rates by Segment

Segment	N	Subscribers	Conversion Rate
Age 18–29	5,273	928	17.6%
Age 30–39	18,089	1,913	10.6%
Age 40–49	11,655	1,063	9.1%
Age 50–59	8,410	785	9.3%
Age 60+	1,784	600	33.6%
Not-Labor-Force	4,505	987	21.9%
White-Collar	26,380	3,141	11.9%
Self-Employed	3,066	310	10.1%
Blue-Collar	10,972	817	7.4%

The model comparison shows that campaign execution is the main driver of predictive performance. Demographics-only modeling reaches an AUC of 0.642, while campaign-only modeling jumps to 0.899. The combined model improves slightly to 0.906, which means demographics add only limited incremental value once behavioral and contact variables are included.

Execution Over Targeting: The Primacy of Campaign Variables (RQ2)

Table 3: Logistic Regression Model Performance Comparison (AUC)

Model	AUC
Model 1: Demographics only	0.642
Model 2: Campaign Execution only	0.899
Model 3: Combined	0.906

The strongest predictors in the combined model are seasonal timing and campaign history. March, October, September, December, and June all show much higher odds of subscription than January, and previous success strongly predicts future success. Longer call duration increases response, especially among older customers, while higher contact frequency reduces response across all job groups.

Tailoring the Approach: Interaction Effects Across Segments (RQ3)

While demographics lack standalone predictive power, they can **moderate** the effectiveness of campaign tactics. We tested two important interactions:

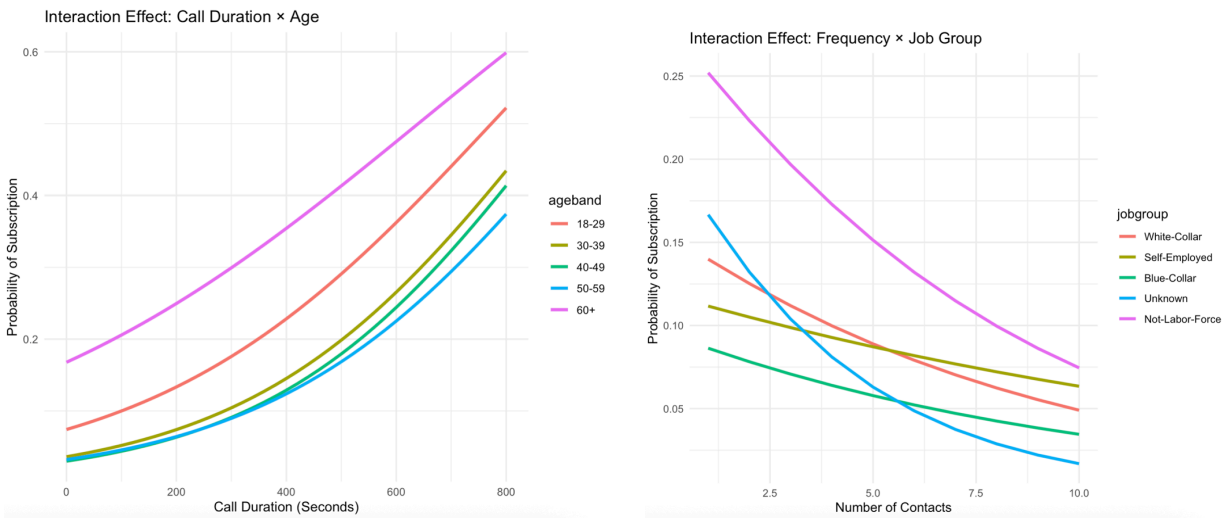


Figure 1: Interaction Effect A - Call Duration × Age

Figure 2: Interaction B - Frequency × Job Group

RQ3 shows that campaign tactics do not work uniformly across customer groups. Longer calls increase subscription probability for all ages, but the effect is much stronger for the 60+ group, while repeated contacts reduce response across all job groups and especially harm blue-collar and unknown segments. The conclusion is that the bank should use segment-specific calling rules: longer, consultative calls for older customers, and tighter limits on contact frequency for working-age and lower-engagement groups.

The Financial Reality: Where We Lose Money (RQ4):

Table 4: Strategic Decision Ranking by EMV

Job Group	Age Band	N	Conv. Rate	Avg. Contacts	Avg. Balance	Cost	EMV (Standard)	EMV (Adjusted)
Not-Labor-Force	60+	1,108	37.5%	2.11	\$2,500.87	\$84.48	+\$43.37	+\$15.23
Not-Labor-Force	18-29	880	28.4%	2.21	\$1,304.87	\$116.58	+\$23.70	-\$14.58
White-Collar	60+	382	29.8%	2.63	\$2,353.59	\$132.11	+\$20.26	-\$4.31
Not-Labor-Force	30-39	700	16.4%	2.56	\$1,425.42	\$233.48	-\$5.50	-\$26.65
Blue-Collar	60+	199	20.1%	2.51	\$1,808.21	\$187.12	+\$2.59	-\$19.44
Self-Employed	18-29	277	17.3%	2.39	\$930.81	\$206.56	-\$1.14	-\$27.73
White-Collar	18-29	2,912	16.8%	2.50	\$1,008.03	\$223.01	-\$3.86	-\$28.98
White-Collar	30-39	11,908	11.6%	3.17	\$1,255.49	\$367.61	-\$19.41	-\$35.30

Note: Top 8 groups sorted by the EMV (Standard). N = Segment size. Conv. Rate = Proportion subscribing. Avg. Contacts = Mean contact attempts per customer. Avg. Balance = Mean account balance for segment. Acq. Cost = (Avg Contacts / Conv Rate) × \$15. EMV (Standard) = (Conv Rate × \$200) - (Avg Contacts × \$15). EMV (Adjusted) = (Conv Rate × Avg Balance × 0.05) - (Avg Contacts × \$15).

The EMV analysis changes the practical conclusion. Although several segments look attractive under a standard \$200 lifetime-value assumption, only Not-Labor-Force 60+ remains profitable after using balance-adjusted EMV. This matters because some groups respond well but do not hold enough balance to justify expensive phone outreach. In other words, the bank should not optimize for conversion rate alone; it should optimize for profitable conversion.

6. Recommendations

The bank should concentrate its outbound calling efforts on the Not-Labor-Force segment aged 60+, as this is the only group that remains consistently profitable under both the standard and balance-adjusted EMV models, while also showing strong responsiveness to longer, more consultative calls. A tailored “premium senior” calling approach would therefore be appropriate, with agents using a relationship-oriented script, allowing sufficient call time, focusing on reassurance about the safety of funds, and prioritising experienced staff for these interactions. At the same time, EMV should become the primary KPI for campaign targeting, since response rates alone can give a misleading picture of performance by overstating the value of some high-response but low-profit segments.

Customers in groups such as White-Collar and Self-Employed aged 18–39 and 30–39, who may respond reasonably well but generate negative adjusted EMV, should be shifted toward lower-cost digital channels such as email, SMS, and in-app messaging, while still keeping a call-back option available for those who request direct contact. The bank should also introduce a quarterly review process to update conversion rates, balances, and call costs so that segment rankings remain aligned with changing behaviour and market conditions. Finally, campaign scheduling should take advantage of the strongest seasonal periods, particularly March, September, and October, by testing small pilot campaigns with messages tailored to those months.

7. Conclusion

In line with the life cycle wealth theory, older and “*Not-In-The-Labor-Force*” exhibited the highest conversion rates, reflecting stronger balance positions and greater preference for more secure investments. By contrast, the large 30-39 segment had generated many



subscribers mainly due to scale rather than a strong individual likelihood of conversion. Consistent with prior findings in the Portuguese dataset, the logistic models show that the campaign variables (e.g. call duration, contact history) explain most of the variation in customer response, with demographic characteristics adding only limited value once the behavioral factors are controlled for. This further demonstrates that campaign tactics are not universally effective. So, while longer, consultative calls significantly improve outcomes for older people, they simultaneously erode responsiveness amongst working age and blue-collar groups. This confirms that a uniform contact strategy is inefficient and that campaign design must be segment specific.

The EMV framework converts these behavioral patterns into explicit financial decision rules. Only in the “*Not-In-The-Labor-Force*” 60+ segment remains consistently profitable under both standard and balance adjusted EMV, while several large white-collar groups were value destroying despite having acceptable response rates. The findings directly support the literature’s warning that response optimized targeting can misallocate marketing budgets. High predicted conversion alone cannot guarantee commercial viability once acquisition costs and customer balance are jointly considered.

Strategically, the bank should concentrate outbound phone calls on the highest EMV age/job segments particularly in the “*Not-In-Labor-Force*”, where longer calls are justified by both higher response and stronger balances. Large but low EMV groups, such as white collar, should be migrated to lower cost digital channels. Similarly, call frequency and duration for working age and blue collar segment should be capped. The EMV calculations should be refreshed regularly to maintain aligned with current profitability rather than being just a single campaign snapshot.

